

LESSON 7.8

WRITING EQUATIONS OF LINEAR RELATIONSHIPS



LESSON INTRODUCTION

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph or an equation in slope-intercept form.

LEARNING TARGETS

- I can determine and interpret the slope and y-intercept of a two-variable linear equation from a table of values in a real-world context.
- I can determine and interpret the slope and y-intercept of a two-variable linear equation from a written description in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

WORKING TOWARD

MA.912.AR.2.5 Solve and graph mathematical and real-world problems that are modeled with linear functions. Interpret key features and determine constraints in terms of the context.

ESSENTIAL QUESTIONS

- How do you determine and interpret the slope and y-intercept of a two-variable linear equation from a table of values in a real-world context?
- How do you determine and interpret the slope and y-intercept of a two-variable linear equation from a written description of a real-world context?

LESSON NARRATIVE

Students will determine and interpret the slope and y-intercept in context when given either a table of values or a written description. Students will use their knowledge of how to find slope using a table or points as well as determining the y-intercept using the slope found from two points. They will then interpret these values in each real-world context.

COMPONENT SUMMARY

7.7.1 WARM-UP (~5 MIN.)

Interpret the slope and y-intercept of an equation on a Florida-based context

7.7.3 GUIDED PRACTICE (10 MIN.)

Practice determining and interpreting the slope and y-intercept from tables and written descriptions in context with a partner

7.7.5 WRAP-UP (~5 MIN.)

Demonstrate understanding of determining and interpreting the slope and y-intercept from a written description in context

7.7.2 GUIDED INSTRUCTION (10-15 MIN.)

Learn how to determine and interpret the slope and y-intercept from tables and written descriptions in context

7.7.4 YOUR TURN! (10 MIN.)

Independently practice determining and interpreting the slope and y-intercept from tables and written descriptions in context

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Slope
- Rate of change
- y-intercept
- Linear relationship

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to find the y-intercept from a table of values when it is not included in the table.
- Students may have a difficult time finding the y-intercept in a written description if they do not understand it is the initial value in the situation.
- Students may struggle to interpret specific contexts if they are not familiar with the contexts.

THINGS TO CONSIDER

- Consider explicitly spiraling content by assigning students to calculate the slope using different values in the table to reiterate that it does not matter which two points are used.
- Consider allowing students to use a calculator to perform calculations. This will allow them to focus on interpreting the values.
- Provide students the opportunity to read and discuss each context prior to performing calculations and interpreting to ensure understanding of each context.

LITERACY AND LANGUAGE ROUTINES

Three Reads

7.7.2 Guided Instruction and 7.7.3 Guided practice provide opportunities to use this structure. Consider applying the Three Reads routine as students make sense of each unique context. This routine supports reading comprehension and supports students in negotiating what information is relevant from a text when discussing with their partner.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.7.1 Real-World Contexts**

Students will be asked to interpret slopes and y-intercepts of linear representations in real-world contexts. Given a table of values or a written description, students will interpret their findings according to the context. For example, in 7.7.5 Wrap-Up, students must determine the slope and y-intercept from the written description and then interpret their meanings to the context.

DIFFERENTIATE INSTRUCTION

Both 7.7.3 Guided Practice and the 7.7.4 Your Turn! components provide students the opportunities to determine and interpret the slope and y-intercept using both a table of values and a written description. Consider combining these two components and allowing students to choose which questions to complete. For example, there are two questions where the student is given a table of values. Allow the student to choose which one to complete. Consider engaging students with the Think-Pair-Share routine in 7.7.3 Guided Practice. The discussions will provide auditory entry points.

7.8.1 WARM-UP: THE NEED FOR SPEED**ENRICHMENT**

- How did you determine the average speed using the information given?
- If both planes were graphed on a coordinate plane with the same y-intercept, which would produce a steeper line? Why?

**7.8.1 Warm-Up: The Need for Speed**

The fastest jet in the world is the Lockheed SR-71 Blackbird. On its last flight, March 6, 1990, a record speed was recorded by flying from Los Angeles, CA to Washington, D.C. in 1 hour, 4 minutes, and 20 seconds.



A typical passenger airplane completed the same route in about 4 hours and 45 minutes.



The flight distance between Los Angeles, CA and Washington, D.C. is 2,280 miles.

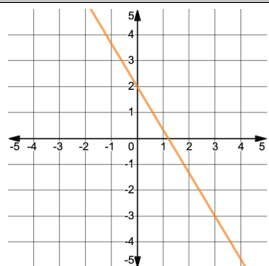
1. What was the Blackbird's average speed when it set the record speed? Round your answer to the hundredths place.

$$\frac{2280 \text{ mi}}{1 \text{ hr } 4 \text{ min } 20 \text{ s}} \times \frac{60 \text{ min}}{1 \text{ hr}} \approx 2126.4 \text{ mph}$$



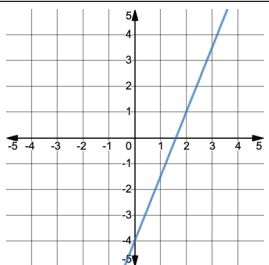
7.8.2 Guided Practice: Writing the Equation of a Linear Relationship

- For each linear relationship shown, work with your partner to determine the slope and y-intercept of the line.

Linear Relationship A	Slope	y-intercept
	$-\frac{5}{3}$	$(0, 2)$ or 2

Linear Relationship B					Slope	y-intercept										
<table><tr><td>x</td><td>-10</td><td>0</td><td>3</td><td>4.5</td></tr><tr><td>y</td><td>2</td><td>6</td><td>7.2</td><td>7.8</td></tr></table>					x	-10	0	3	4.5	y	2	6	7.2	7.8	$\frac{2}{5}$	$(0, 6)$ or 6
x	-10	0	3	4.5												
y	2	6	7.2	7.8												

$$\frac{6 - 2}{0 - (-10)} = \frac{4 - 2}{10 - 2} = \frac{2}{5}$$

Linear Relationship C	Slope	y-intercept
	$\frac{5}{2}$	$(0, -4)$ or -4



Slope-intercept form of linear equations is written in the form $y = mx + b$, where m is the slope, or rate of change, of the graph, and b is the y-intercept of the graph.

7.8.2 GUIDED PRACTICE: WRITING THE EQUATION OF A LINEAR RELATIONSHIP

ENRICHMENT

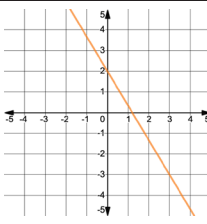
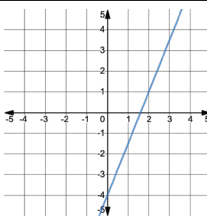
- How do you find the slope from each representation?
- How do you find the y-intercept from each representation?

7.8.2 GUIDED PRACTICE: WRITING THE EQUATION OF A LINEAR RELATIONSHIP (CONTINUED)

ENRICHMENT

- What is the slope-intercept form of a line?
- How do you know whether to use + or – in the equation?

2. Use the slope and y-intercept for each linear relationship to write the equation of each line in slope-intercept form.

Linear Relationship	Equation										
	$y = -\frac{5}{3}x + 2$										
<table border="1" data-bbox="958 525 1201 596"><tr><td>x</td><td>-10</td><td>0</td><td>3</td><td>4.5</td></tr><tr><td>y</td><td>2</td><td>6</td><td>7.2</td><td>7.8</td></tr></table>	x	-10	0	3	4.5	y	2	6	7.2	7.8	$y = \frac{2}{5}x + 6$
x	-10	0	3	4.5							
y	2	6	7.2	7.8							
	$y = \frac{5}{2}x - 4$										



7.8.3 Activity: Gallery Walk

Several students wrote equations to represent linear relationships given as tables or graphs, but each student made at least one error.

With your partner or small group, complete the table by analyzing each student's work to identify their error(s). Then write the correct equation for each linear relationship.

Student's Name	What error(s) did they make?	Correct Equation
Sophia	<i>Switched Δx and Δy in slope and used the x-intercept instead of y-intercept.</i>	$y = -\frac{2}{5}x + 1$
Austin	<i>Didn't realize the first y-value wasn't for $x = 0$</i>	$y = 3.4x - 2.7$
Caleb	<i>Used the wrong sign for the y-intercept and did not include x in equation</i>	$y = \frac{3}{2}x - 4$
Hailey	<i>Only looked at Δy but did not divide by Δx</i>	$y = -0.75x + 7$

7.8.3 ACTIVITY: GALLERY WALK

TEACHER MOVES

This activity provides an opportunity for deep discussion regarding common errors. Help lead students to have those discussions with their group or partner and encourage students to discuss how they might avoid those same mistakes in their work. Monitor for strategies discussed by the students to share with the entire class.

ENRICHMENT

- What did the student do correctly?
- Where did the student make errors?
- What was the mistake, and why do you think the student made it?
- What are some strategies you could use to avoid making similar errors?

7.8.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider using the Think-Pair-Share routine to promote discussion for students who learn best from hearing and verbalizing their ideas. Possibly provide sentence starters to guide the conversation.

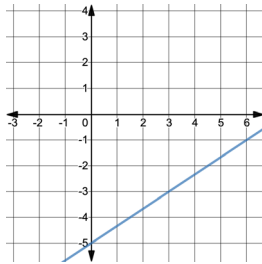
- “The slope is _____ because...”
- “The y-intercept is _____ because...”
- “The equation is _____ because...”
- “I agree with you because _____...”
- “I disagree with you because _____...”



7.8.4 Your Turn!

1. Write the equation of each line in slope-intercept form.

Linear Relationship					Equation
x	-4	-2	0	2	$y = -\frac{3}{8}x + \frac{1}{2}$
y	2	$\frac{5}{4}$	$\frac{1}{2}$	$-\frac{1}{4}$	

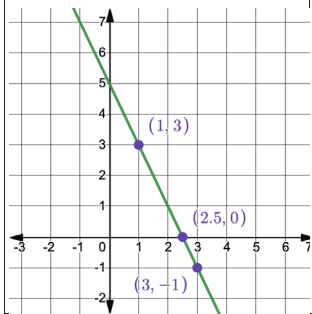
	$y = \frac{2}{3}x - 5$

$$\frac{\frac{1}{2} - 2}{0 - (-4)} = \frac{-1.5}{4} = -0.375$$



7.8.5 Wrap-Up: Ticket Out the Door

1. The graph and table each represent linear relationships. Write the equation of each line in slope-intercept form.

Linear Relationship	Equation														
	$y = -2x + 5$														
<table border="1"><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>-4</td><td>-5</td></tr><tr><td>0</td><td>-3</td></tr><tr><td>1</td><td>-2.5</td></tr><tr><td>3</td><td>-1.5</td></tr><tr><td>6</td><td>0</td></tr><tr><td>10</td><td>2</td></tr></tbody></table>	x	y	-4	-5	0	-3	1	-2.5	3	-1.5	6	0	10	2	$y = \frac{1}{2}x - 3$
x	y														
-4	-5														
0	-3														
1	-2.5														
3	-1.5														
6	0														
10	2														

$$\frac{-2.5 - (-5)}{1 - (-4)} = \frac{2.5}{5} = \frac{1}{2}$$

7.8.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Discuss any questions the students still have regarding the content. Consider making a list of the questions on the board or poster to come back to and address later.